

On Approximate Horn Minimization

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Abstract

The minimization problem for Horn formulas is to find a Horn formula equivalent to a given Horn formula, using a minimum number of clauses. A $2^{\log^{1-\epsilon}(n)}$ -inapproximability result is proven, which is the *first* non-trivial inapproximability result for this problem. We also consider several other versions of Horn minimization. The more general version which allows for the introduction of new variables is known to be too strong as its equivalence problem is co-NP-complete. Therefore, we propose a variant called *Steiner-minimization*, which allows for the introduction of new variables in a restricted manner. Steiner-minimization of Horn formulas is shown to be MAX-SNP-hard. In the positive direction, a $o(n)$, namely, $O(n \log \log n / (\log n)^{1/4})$ -approximation algorithm is given for the Steiner-minimization of definite Horn formulas. The algorithm is based on a new result in algorithmic extremal graph theory, on partitioning bipartite graphs into complete bipartite graphs, which may be of independent interest. Inapproximability results and approximation algorithms are also given for restricted versions of Horn minimization, where only clauses present in the original formula may be used.

1 Introduction

The CNF minimization problem is to find a shortest CNF expression equivalent to a given expression. This problem has been studied, in different versions for many decades in switching theory, computer science and engineering, and it is still the topic of active research, both in complexity theory and circuit design. Umans [42, 43] showed Σ_p^2 -completeness and $O(n^{1-\epsilon})$ -inapproximability result for this problem. Horn minimization is the special case of CNF minimization for Horn formulas. Horn formulas are conjunctions of *Horn clauses*, *i.e.*, of disjunctions containing *at most one* unnegated variable, and therefore can also be written as implications. For instance, $\bar{a} \vee \bar{b} \vee c$ is a Horn clause which can also be written as $a, b \rightarrow c$.

Horn formulas are an expressive and tractable fragment of propositional logic, and therefore provide a basic framework for knowledge representation and reasoning [36]. Horn formulas provide, for example, a natural framework for representing systems of rules for expert systems. An interesting potential new applica-

tion area for Horn formulas is the automated, interactive development of large-scale knowledge bases of commonsense knowledge (see [37] for the description of such a project). This application has algorithmic aspects involving knowledge representation, reasoning, learning and knowledge update. A model incorporating these aspects, called *Knowledge Base Learning (KnowBLe)* is formulated in [29] (see also [28, 30]) for related work). Efficient algorithms for approximate Horn minimization would be useful in these applications.

Satisfiability of Horn formulas can be decided in linear time and the equivalence of Horn formulas can be decided in polynomial time [25]. Thus Horn minimization is expected to be easier than CNF minimization. Horn minimization was shown to be NP-complete by Hammer and Kogan [20] if the number of literals is to be minimized, and by Boros and Čepek [8] if the number of clauses is to be minimized. On the positive side, Hammer and Kogan [21] gave a polynomial algorithm for minimizing quasi-acyclic Horn formulas, which include both acyclic and 2-Horn formulas. It was also shown in [20] that there is an efficient $(n - 1)$ -approximation algorithm for general Horn minimization, where n is the *number of different variables* in the formula (not the number of variable occurrences). As noted in [18], such an algorithm is also provided by the Horn formula learning algorithm of [3].

1.1 Contributions of this paper

First, in Theorem 3 we prove a $2^{\log^{1-\epsilon}(n)}$ -inapproximability result for Horn minimization assuming $\text{NP} \not\subseteq \text{DTIME}(n^{\text{polylog}(n)})$ via a reduction from the MINREP problem [35, 26]. This seems to be the *first* non-trivial inapproximability result for this problem. We next consider several other versions of the Horn minimization problem. Depending on the application, different versions may be relevant and thus of interest for exploration of their approximability properties. This aspect of Horn minimization does not seem to have been considered before.

It may be possible to *add new variables* in order to compress the formula. For example, the formula

$$\varphi = \bigwedge_{i=1}^n \bigwedge_{j=1}^n (x_i \rightarrow y_j) \quad (1)$$

having n^2 clauses can be compressed to the $2n$ clause formula

$$\psi = \bigwedge_{i=1}^n (x_i \rightarrow z) \wedge \bigwedge_{j=1}^n (z \rightarrow y_j), \quad (2)$$

where z is a new variable. Note that φ and ψ are clearly *not* equivalent, *e.g.*, φ does not depend on z , while ψ does. On the other hand, φ and ψ are equivalent in the sense that they both imply the same set of clauses over the original variables $x_1, \dots, x_n, y_1, \dots, y_n$. Thus, in terms of the knowledge base application, the new variable z can be thought of as being *internal* to the knowledge base and invisible to the user. Flögel *et al.* [17] showed that deciding the equivalence of

such extended Horn formulas is co-NP-complete. This is bad news, as it suggests that the extended version is too expressive and therefore intractable⁴.

On the other hand, notice that in the example above the new variable is added in a rather *restricted manner*. Formula (1) can be thought of as a complete directed bipartite graph with parts $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$. Formula (2), then, represents the operation of adding a new node z in the middle, with edges from the x_i 's to z and from z to the y_j 's. The two graphs have the same reachability relations as far as the original nodes are concerned. Using the similarity to Steiner problems where new points may be added [22], we refer to this operation as a *Steiner extension* (a formal definition appears in Section 5). As we observe, in contrast to general extensions, the equivalence of Steiner extensions *can* be decided efficiently (Corollary 1). Thus this type of extension could be considered as a tractable alternative in the applications mentioned. The *Steiner minimization* problem for Horn formulas is then to find an equivalent Steiner-extended Horn formula, for a given Horn formula, with a minimum number of clauses. We show in Theorem 5 that this problem is MAX-SNP-hard. On the other hand, in Theorem 6 we prove that there is an efficient $O(n \log \log n / (\log n)^{1/4})$ -approximation algorithm for this problem, where n is the number of variables in the original formula. This is the *first* approximation algorithm for Horn minimization with a $o(n)$ approximation guarantee.

The algorithm for Steiner minimization makes use of an algorithmic result on the *partition of bipartite graphs into complete bipartite graphs* (i.e., *bicliques*), which may be of interest on its own. It was shown by Chung, Erdős and Spencer [13] and Bublitx [10] that the edges of every n -vertex graph can be partitioned into complete bipartite graphs such that the sum of the number of vertices in these bipartite graphs⁵ is $O(n^2 / \log n)$, and this is asymptotically the best possible. Tuza [41] gave an analogous result for bipartite graphs. These results are based on a counting argument due to Kővári, Sós and Turán [27], which shows that sufficiently dense graphs contain large complete bipartite subgraphs, and thus are non-constructive. Kirchner [23] considered the problem of finding an algorithmic version, and gave an efficient algorithm to find complete balanced bipartite graphs of size $\Omega(\sqrt{\log n})$ in dense graphs. In a previous paper [33] we improved this to the optimal $\Omega(\log n)$, and as a corollary, showed that partitions proved to *exist* in [10, 13] can also be *found efficiently*⁶.

⁴ Introducing new variables has been considered earlier, going back to Tseitin [40].

While there are many exponential lower bounds for resolution proofs (see, e.g., [14]), complexity-theoretic results suggest that proving such results for extended resolution is much harder [15].

⁵ Note that the complexity of a partition is not measured by the number of graphs in it, but by a different measure, which comes from circuit complexity [38].

⁶ Extremal combinatorics provides results on the existence of substructures. The results of [23] and [33] can be viewed as *algorithmic extremal combinatorics* as they also give efficient algorithms to actually find such substructures. Previous results in this direction are given in [2]. The results of [2] apply to dense graphs and find substructures of constant size, while here we have to handle sparser graphs as well and to find substructures of nonconstant size.

In this paper we give an algorithmic version for the bipartite case. It turns out that Tuza's result [41] is not quite correct, as it fails for very unbalanced bipartite graphs, such as a star. We show in Theorem 1 that the edges of every bipartite graph with sides a and b , where $a \geq b$, can be partitioned into complete bipartite graphs such that the sum of the number of vertices of these graphs is $O((ab/\log a) + a \log b)$, and we give an efficient algorithm to find such a partition.

We also consider *restricted* versions of the Horn minimization problem, where one is restricted to use clauses from the original formula. Such a restriction may be justified in applications where the particular rules, provided by an expert, are supposed to be meaningful and thus cannot be replaced. The goal is to eliminate redundant rules. Modifying the construction of Theorem 3, in Theorem 7 we prove $2^{\log^{1-\epsilon}(n)}$ -inapproximability for the restricted case, which holds even if the input formula has clauses of size at most 3.

One may want to optimize a Horn formula in the restricted sense either by *minimizing the number of rules left*, or by *maximizing the number of rules removed*. The two versions may differ in their approximability (cf. the maximum independent set and the minimum vertex cover problems for graphs). As (1) suggests, Horn formulas with clauses of the form $x \rightarrow y$ correspond to directed graphs. For such formulas, optimization corresponds to *transitive reduction* problems for directed graphs. Thus approximation algorithms for these directed graph problems (in both versions) may be applied for Horn formulas. Examples of this connection are given in Theorem 8.

The rest of the paper is organized as follows. Bipartite graph decompositions are discussed in Section 3. Section 4 contains the inapproximability result for Horn minimization. Horn minimization with new variables is discussed in Section 5, and restricted Horn minimization in Section 6. *Most proofs omitted due to lack of space and several proof outlines appear in the Appendix.*

2 Preliminaries

A *clause* is a disjunction of literals. A *Horn clause* is a clause with at most one unnegated variable. A *definite* Horn clause has exactly one unnegated variable, called its *head*; the other variables form its *body*. A *negative* clause consists of only negated variables. The *size* of a clause is the number of its literals. A clause of size 1 (resp., 2) is a *unit* (resp., *binary*) clause. A *(definite) Horn formula* is a conjunction of (definite) Horn clauses⁷. The *size* of a formula is the number of its clauses. A *k-Horn formula* is a Horn formula with clauses of size at most k .

A clause C is an *implicate* of a formula φ if every truth assignment satisfying φ also satisfies C . An implicate is a *prime* implicate if none of its proper sub-clauses is an implicate. Deciding whether a *definite Horn* clause C is an implicate of a *definite Horn* formula φ can be decided by a simple and well-known marking procedure often called *forward chaining*. The procedure begins by marking

⁷ The extension of the algorithms to the general case will be discussed in the final version of the paper. Note that structural results on Horn formulas show that negative clauses are often easier to handle than definite ones [19].

the variables in the body of C . If every variable in the body of a clause in φ is marked then its head is marked as well. This is repeated as long as new variables get marked. Then it holds that C is an implicate of φ iff its head gets marked.

3 Partitioning and covering bipartite graphs using bicliques

Let $G = G(A, B, E)$ be a bipartite graph with parts A, B of sizes a, b , respectively, and edge set E of size m . We assume w.l.o.g. that $a \geq b$. A bipartite graph is *balanced* if $|A| = |B|$. The complete bipartite graph (or biclique) with parts of size p and q is denoted by $K_{p,q}$.

We consider bicliques $G_i = (A_i, B_i, E_i)$ for $i = 1, \dots, t$ such that $A_i \subseteq A, B_i \subseteq B$, and $(E_1, \dots, E_t)$ is a partition, resp. a cover, of E . The cost of such a decomposition is $\sum_{i=1}^t (|A_i| + |B_i|)$. The problem is to find a decomposition of small cost. The trivial decomposition into single edges has cost $2m \leq 2ab$.

We consider two versions of the problem. In the first version we are interested in finding a partition such that its size is upper bounded by some function of a and b , independent of m .

Theorem 1. *For every bipartite graph G one can find a partition of cost $O\left(\frac{ab}{\ln a} + a \ln b\right)$ in polynomial time.*

The decomposition is found by iteratively finding large bipartite subgraphs. There are two procedures, depending on a carefully chosen notion of density. Let $3 \leq b \leq a, 6a \leq m \leq ab$ and

$$f(a, b, m) = \left\lfloor \frac{\ln a}{\ln(2eab/m)} \right\rfloor;$$

note that the case $b \leq 2$ is trivial.

Lemma 1. *Suppose that $m \geq af(a, b, m)$. Then there is a polynomial time algorithm that finds a $K_{q,q}$ in G with $q = f(a, b, m)$.*

Lemma 2. *Suppose that $m < af(a, b, m)$. Then there is a polynomial time algorithm that finds a $K_{q,q}$ in G with $q = \lfloor m/a \rfloor$.*

Remark 1. If G is a star then $b = 1$ and the optimal decomposition has cost $a + 1$, hence the upper bound $ab/\ln a$ claimed in [41] does not hold, and an additional term (or some other modification) is needed. It is open whether the term $a \ln b$ can be improved.

In the second version we are interested in finding a partition (resp., cover) of minimal cost. For technical reasons, we use a slightly different cost function here (using this cost function would not change anything in the previous result). The size of $K_{p,q}$ is $p \cdot q$ and the modified cost of $K_{p,q}$ is

$$\text{cost}'(K_{p,q}) = \begin{cases} p \cdot q & \text{if } p = 1 \text{ or } q = 1 \\ p + q & \text{otherwise} \end{cases}$$

We define the LINEAR-COST-BICLIQUE-COVER (resp., LINEAR-COST-BICLIQUE-PARTITION) problem as follows: given a bipartite graph $G = (A, B, E)$, cover (resp., partition) its edges with bicliques of minimum total *modified* cost. The minimization of the number of bicliques in a cover was shown to be *NP*-complete by Orlin [34]. We prove the following result by giving an approximation-preserving reduction from the maximum independent set problem for 3-regular graphs.

Theorem 2. *Assuming $P \neq NP$, LINEAR-COST-BICLIQUE-COVER and LINEAR-COST-BICLIQUE-PARTITION cannot be approximated in polynomial time within an approximation ratio of $1 + (1/1138)$ (≈ 1.0008) even if the input graph has no biclique of size more than 6.*

4 Inapproximability

Theorem 3. *For any fixed $0 < \epsilon < 1$, unless $NP \subseteq DTIME(n^{\text{polylog}(n)})$, the Horn minimization problem for definite Horn formulas is $2^{\log^{1-\epsilon} n}$ -inapproximable.*

The reduction is from the MINREP problem [26]. An instance M is given by a bipartite graph $G = (A, B, E)$ with $|E| = m$, a partition of A into $|A|/\alpha$ equal-size subsets $A_1, A_2, \dots, A_\alpha$ and a partition of B into $|B|/\beta$ equal-size subsets $B_1, B_2, \dots, B_\beta$. One can define a natural bipartite super-graph H in the following manner. H has a super-vertex for every A_i and B_j . There is a super-edge between A_i and B_j if and only if there exists $u \in A_i$ and $v \in B_j$ such that (u, v) is an edge of G . Let the number of super-edges be p . A pair of nodes u and v *witnesses* a super-edge (A_i, B_j) provided $u \in A_i$, $v \in B_j$ and the edge (u, v) exists in G . A set of nodes S of G witnesses a super-edge if and only if there exists at least one pair of nodes in S that witnesses the super-edge. The goal of MINREP is to find $A' \subseteq A$ and $B' \subseteq B$ such that $A' \cup B'$ witnesses every super-edge of H and $|A'| + |B'|$ is as small as possible. The *size* of an optimal solution is denoted by $OPT(M)$. Let $s = |A| + |B|$.

Consider an instance M of MINREP. Let t be a sufficiently large positive integer to be fixed later. We construct a definite Horn formula φ . For simplicity and with an abuse of notation, some variables in φ are denoted as the corresponding objects (vertices and super-edges) in the MINREP instance. The formula φ contains *amplification* variables $x_1, \dots, x_t$, *node* variables u for every vertex u in $A \cup B$ and *super-edge variables* e for every super-edge e in H . The clauses of φ belong to the following groups:

amplification clauses: there is a clause $x_i \rightarrow u$ for every $i \in \{1, \dots, t\}$ and for every $u \in A \cup B$.

witness clauses: there is a clause $u, v \rightarrow e$ for every super-edge e of H and for every pair of nodes $u \in A$ and $v \in B$ witnessing e .

feedback clauses: there is a clause $e_1, \dots, e_p \rightarrow u$ for every $u \in A \cup B$, where $e_1, e_2, \dots, e_p$ are the super-edges of H .

As φ is definite, all its prime implicates are definite. Also, as φ consists of non-unit definite clauses, all its prime implicates are non-unit (the all-zero vector satisfies φ and falsifies all unnegated variables). For a further analysis of the prime implicates of φ , we make use of forward chaining.

Lemma 3. *Let x be an amplification variable. Then the prime implicates containing x are clauses of the form $x \rightarrow v$, where v is a node or super-edge variable.*

Lemma 4. *Let U be a set of node variables such that U is not a solution to MINREP, and U' be the set of super-edge variables witnessed by U . Then every implicate with body contained in $U \cup U'$ has head in U' .*

Lemma 5. *Let x be an amplification variable and let ψ be a prime and irredundant Horn formula equivalent to φ . Then ψ has at least $OPT(M)/3$ clauses containing x .*

Based on these lemmas one can prove that the reduction is gap-preserving.

Lemma 6 (Gap preserving reduction lemma).

- (a) *If $OPT(M) = \alpha + \beta$, then $OPT(\varphi) \leq t \cdot (\alpha + \beta) + m + s$.*
- (b) *If $OPT(M) \geq (\alpha + \beta) \cdot 2^{\log^{1-\epsilon} s}$ then, $OPT(\varphi) \geq t(\alpha + \beta) \cdot 2^{\log^{1-\epsilon} s}/3$.*

Our result now follows from the known approximation hardness of MINREP [35, 26].

5 Formulas with new variables

In this section we consider versions of the Horn minimization problem where one can introduce new variables in order to compress the formula.

5.1 General extensions

First we consider the general version where there is no restriction on the way the new variables are introduced

Definition 1 (Generalized equivalence). [17] *Let X be a set of variables. Formulas φ and ψ are X -equivalent if for every clause C over X it holds that $\varphi \models C$ iff $\psi \models C$.*

Consider the set of variables $X = \{x_1, \dots, x_n, y_1, \dots, y_n, u\}$ and the 2^n -clause Horn formula $\varphi = \bigwedge (v_1, \dots, v_n \rightarrow u)$ where $v_i \in \{x_i, y_i\}$ for $i = 1, \dots, n$, and the conjunction includes all possible such selections. As no resolutions can be performed, it follows that all the prime implicates of φ are the clauses themselves. Let now $\{z_1, \dots, z_n\}$ be new variables. Then the $(2n + 1)$ -clause Horn formula $\psi = (z_1, \dots, z_n \rightarrow u) \wedge \bigwedge_{i=1}^n (x_i \rightarrow z_i) \wedge (y_i \rightarrow z_i)$ is X -equivalent to φ . Thus the introduction of new variables can lead to an exponential compression in size. It was shown by [17] that generalized equivalence is too general for applications, as deciding it for Horn formulas is hard.

Theorem 4. [17] *Generalized equivalence of definite Horn formulas is co-NP-complete.*

5.2 Steiner extension

The proof of Theorem 4 shows that generalized equivalence is already hard if new variables are introduced in a rather restricted manner. This gives a motivation to consider even more stringent restrictions on the introduction of new variables.

Definition 2 (Steiner extension). *Let φ be a Horn formula and X be a subset of its variables. Then φ is a Steiner extension over X if every variable not in X occurs in φ either as a head, or as a single body variable in a binary definite clause having its head in X .*

The corresponding notion of equivalence is the following.

Definition 3 (Steiner equivalence). *Let X be a set of variables. Horn formulas φ and ψ are Steiner X -equivalent if*

- φ and ψ are X -equivalent,
- both φ and ψ are Steiner extensions over X .

Note that the Horn formulas in (1) and (2) above are both Steiner extensions over $X = \{x_1, \dots, x_n, y_1, \dots, y_n\}$ (the first formula vacuously so), and they are Steiner X -equivalent. On the other hand, the example above is not a Steiner extension as additional variables occur in the body of a non-binary clause. In contrast to Theorem 4, Steiner equivalence can be decided efficiently.

Proposition 1. *There is a polynomial algorithm which, given a Steiner extension φ over X , computes a Steiner X -equivalent Horn formula $\psi(X)$ such that $\text{size}(\psi) = O(\text{size}(\varphi^2))$.*

Corollary 1. *Steiner equivalence of Horn formulas is in P .*

The minimization problem for Steiner equivalence is the following.

Definition 4 (Steiner minimization of Horn formulas). *Given a Horn formula φ over variables X , find a minimal size Horn formula that is Steiner X -equivalent to φ .*

As a direct consequence of Theorem 2 we obtain the following.

Theorem 5. *Steiner minimization of Horn formulas is MAX-SNP-hard.*

We now show that Steiner minimization of Horn formulas has an efficient approximation algorithm with performance guarantee $o(n)$.

Remark 2. It may be assumed *w.l.o.g.* that Horn formulas to be minimized have no unit clause prime implicates. This holds as every prime representation can be partitioned into those unit clauses and a set of clauses not containing any variable that occurs in a unit clause prime implicate. The second half then can be minimized separately.

Theorem 6. *There is an efficient*

$$O(n \log \log n / (\log n)^{1/4})$$

approximation algorithm for Steiner minimization of definite Horn formulas, where n is the number of variables in the original formula.

The algorithm uses several procedures. It uses previous algorithms listing prime implicants of Horn formulas and for body minimization. It also uses a procedure for the exact minimization of Horn formulas having a short equivalent formula and the bipartite graph partition algorithm of Section 3.

The *prime implicate listing problem* for Horn formulas is to produce a list of all prime implicants of a Horn formula. As the number of prime implicants can be exponential in the size of the original formula, a possible criterion of efficiency is *total polynomial time*, i.e., time polynomial in the combined size of the input and the output. Boros, Crama and Hammer [9] give an algorithm which lists all prime implicants of a Horn formula, in time polynomial in the size of the formula and the number of prime implicants.

Consider the following special case of (standard) Horn minimization.

Problem 1 ($\sqrt{\log n}$ -Horn minimization). Given a Horn formula φ over n variables, find an equivalent minimal size Horn formula of size at most $\sqrt{\log n}$ if such a formula exists, or output ‘none’.

Lemma 7. *The $\sqrt{\log n}$ -Horn minimization problem is polynomial-time solvable.*

A further ingredient of our algorithm is an efficient procedure for body minimization. The *body minimization* problem for Horn formulas asks for an equivalent Horn formula with the minimal number of distinct bodies. While Horn minimization is hard, there *are* efficient algorithms for body minimization. Such algorithms were found in several different contexts, such as functional dependencies for databases [32], directed hypergraphs [4], lattices [11] and computational learning theory[3].

Given a Horn formula χ over a set of variables X , we now describe a construction of a Steiner extension ψ of χ . Let $Bodies(\chi)$ denote the set of bodies in χ , and $Heads(\chi)$ denote the set of heads in χ . Form a bipartite graph $G(\chi)$ with parts $Bodies(\chi)$ and $Heads(\chi)$, adding an edge between a body and a head if the corresponding Horn clause occurs in χ .

Let $G_1, \dots, G_t$ be a decomposition of $G(\chi)$ into complete bipartite graphs⁸. Let the bipartite graphs in the decomposition have parts $A_i \subseteq Bodies(\chi)$ and $B_i \subseteq Heads(\chi)$ for $i = 1, \dots, t$. Introduce new variables $y_1, \dots, y_t$, and let ψ consist of the clauses $b \rightarrow y_i$ and $y_i \rightarrow h$ for every $b \in A_i$ and $h \in B_i$, $i = 1, \dots, t$.

In the following description of the algorithm let *DECOMP* denote the graph partition procedure of Theorem 1 and $\sqrt{\log n}$ -HORN-MIN denote the procedure of Lemma 7.

⁸ The bipartite graphs need not be balanced. Also, for this application it would be sufficient to consider coverings instead of partitions. The result of Section 3 is formulated for balanced partitions in order to give a stronger positive result.

Input: a Horn formula φ
Goal: an equivalent Horn formula χ with a minimal number of bodies
Algorithm: $k = |\text{Bodies}(\chi)|$, $\ell = |\text{Heads}(\chi)|$
if $(k \leq \ell \text{ and } k \leq \sqrt{\log n})$ **then**
if $\sqrt{\log n}$ -HORN-MIN(χ) returns $\psi \neq \text{'none'}$ **then return** ψ
else
return ψ representing $\text{DECOMP}(G(\chi))$

The performance bound of the algorithm follows by considering different cases depending on the value of $\text{OPT}(\varphi)$ and the relationship between k and ℓ .

6 Restricted Horn minimization

A special case of the Horn optimization problem is when every clause of the optimized formula must also be a clause of the new formula. An inapproximability result similar to Theorem 3 holds in the case of restricted Horn minimization as well, *even if we assume that the formula to be minimized is 3-Horn*.

Theorem 7. *For any fixed $0 < \epsilon < 1$, unless $\text{NP} \subseteq \text{DTIME}(n^{\text{polylog}(n)})$, the restricted Horn minimization problem is $2^{\log^{1-\epsilon} n}$ -inapproximable, even for 3-Horn formulas.*

As noted in the introduction, in the case of restricted Horn minimization one can also try to *maximize* the number of deleted clauses. We refer to this problem below as *Horn maximization*. In contrast to Theorem 7, for 2-Horn formulas constant approximation is possible.

Theorem 8.

- (a) *Both the Horn minimization and Horn maximization problems are MAX-SNP-hard for definite 2-Horn formulas without unit clauses.*
- (b) *Restricted Horn minimization for definite 2-Horn formulas admits a 1.5-approximation.*
- (c) *Restricted Horn maximization for definite 2-Horn formulas admits a 2-approximation.*

In view of Remark 2, these results follow from [5, 24, 44] and the correspondence between Horn formulas with binary clauses and directed graphs.

Remark 3. For (a), the best inapproximability constants can be obtained by using a randomized construction of a special class of Boolean satisfiability instances by Berman *et al.*[7] giving an inapproximability constant of $1 + (1/896)$ for the minimization version and $1 + (1/539)$ for the maximization version.

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APPENDIX

A Proof of Theorem 1

We first present proofs of the two lemmas.

A.1 Proof of Lemma 1

Let $r = \lceil qab/m \rceil$. Since $m \geq 6a \geq 6b > 2eb$ we have $q \geq 1$ and therefore $r \geq 1$. Also, $q = f(a, b, m)$ implies that $r \leq b$. So we have $1 \leq r \leq b$. The algorithm works as follows:

- find R , the set of r highest degree vertices in B and let $H = H(R, A)$ be the bipartite graph between R and A ,
- check all q -element subsets Q of R : if vertices in Q have at least q common neighbors in A then output $K_{q,q}$ formed by Q and q such common neighbors and stop.

We need to show that the procedure finds a $K_{q,q}$ and it runs in polynomial time. The first claim follows using the argument of [27]. The bipartite graph H has at least mr/b edges. The number of q -tuples of vertices of R included in the neighborhoods of vertices in A is, using convexity,

$$\sum_{v \in A} \binom{d_H(v)}{q} \geq a \binom{mr/ab}{q} > a \left(\frac{mr}{abq} \right)^q \geq \left(\frac{2re}{q} \right)^q > q \binom{r}{q},$$

where the second inequality is because $r \geq qab/m$ and the third inequality is because $q = f(a, b, m)$. Thus some q -tuple of vertices in R is included in the neighborhood of at least q vertices in A , and so $K_{q,q} \subset H$.

To show that this is a polynomial time algorithm, it suffices to show that $\binom{r}{q}$ is polynomial. Now $\binom{r}{q} < (re/q)^q = e^q (r/q)^q$. Since $q < \ln a$, we have that $e^q < a$. Also

$$(r/q)^q = e^{q \ln(r/q)} < e^{q \ln(2ab/m)} = e^{f(a,b,m) \ln(2ab/m)} < e^{\ln a} = a,$$

where the last inequality follows from the definition of $f(a, b, m)$.

A.2 Proof of Lemma 2

In this case the algorithm is even simpler than that in the proof of Lemma 1, as the same search is performed on G itself. Since $b \geq 3$,

$$q \leq \min \{m/a, f(a, b, m)\}$$

and a similar calculation as in the proof of Lemma 1 gives

$$\sum_{v \in A} \binom{d_G(v)}{q} \geq a \binom{m/a}{q} > a \left(\frac{m}{aq} \right)^q \geq \left(\frac{2be}{q} \right)^q > q \binom{b}{q}.$$

Therefore $K_{q,q}$ is a subgraph of G . To show that the resulting algorithm is polynomial time, we need to show that $\binom{b}{q} < e^q (b/q)^q$ is polynomial. Since $q \leq m/a < f(a, b, m) < \ln a$ we have that $e^q < a$. Also, $q \geq m/2a$ so

$$(b/q)^q = e^{q \ln(b/q)} \leq e^{(m/a) \ln(b/q)} < e^{f(a, b, m) \ln(2ab/m)} \leq a^{\ln(2ab/m) / \ln(2eab/m)} < a.$$

A.3 Completing the proof

The decomposition algorithm calls the procedures described in the lemmas repeatedly, and deletes edges contained in the new subgraph. When the number of edges drops below $6a$, each remaining edge is included as a singleton.

The upper bound for the cost of the decomposition is given using an argument similar to [41]. A bipartite subgraph $K_{q,q}$ adds $2q$ to the complexity of the decomposition: distribute this charge among the q^2 edges of $K_{q,q}$ by assigning a weight $2/q$ to each edge. Note that the density condition distinguishing between the two procedures has m on both sides, but a simple calculation shows that applications of Lemma 1 precede those of Lemma 2.

If Lemma 1 is applied and the current number of edges is between $ab/(s+1)$ and ab/s for some s , then each edge gets weight $\Theta(\ln s / \ln a)$. There are $\Theta(ab/s^2)$ edges in this range. Hence, up to order of magnitude, the total weight is at most

$$\sum_{s=1}^{\infty} \frac{ab}{s^2} \cdot \frac{\ln s}{\ln a} = O\left(\frac{ab}{\ln a}\right).$$

If Lemma 2 is applied and the current number of edges is between $ab/(s+1)$ and ab/s for some s , then each edge gets weight $\Theta(s/b)$. Again, there are $\Theta(ab/s^2)$ edges in this range. By our assumption on the number of edges, it also holds that $s \leq b$. Hence, up to order of magnitude, the total weight is at most

$$\sum_{s=1}^b \frac{ab}{s^2} \cdot \frac{s}{b} = O(a \ln b).$$

B Proof of Theorem 2

Let 3-MIS be the problem of finding a maximal independent set in a 3-regular graph, and let 3-SET-PACKING be the following problem: given a collection $\mathcal{S}$ of 3-element subsets of an universe $\mathcal{U}$ such that every element $u \in \mathcal{U}$ appears in exactly two sets of $\mathcal{S}$ and for any two sets $S, S' \in \mathcal{S}$ we have $|S \cap S'| \leq 1$, find a maximal subfamily of pairwise disjoint sets in $\mathcal{S}$. Then 3-MIS reduces to 3-SET-PACKING by considering 3-element edge sets incident to a single vertex.

Theorem 9. [6] (a) *For every constant $\varepsilon > 0$, it is NP-hard to decide if an instance of 3-MIS with $284n$ nodes has the maximum size of independent set above $(140 - \varepsilon)n$ or below $(139 + \varepsilon)n$.*

(b) *For every constant $\varepsilon > 0$, it is NP-hard to decide if an instance of 3-SET-PACKING with $(3 \times 284n)/2 = 426n$ elements in $\mathcal{U}$ and $284n$ sets in $\mathcal{S}$ has the maximum size set packing above $(140 - \varepsilon)n$ or below $(139 + \varepsilon)n$.*

Consider an instance $(\mathcal{U}, \mathcal{S})$ of 3-SET-PACKING as described in Theorem 9. We generate an instance $G = (A, B, E)$ of LINEAR-COST-BICLIQUE-COVER as follows:

- there is a vertex $a_u \in A$ for every element $u \in \mathcal{U}$ and a vertex $b_S \in B$ for every set $S \in \mathcal{S}$; there is also one additional vertex $b^* \in B$,
- for every element u and set S such that $u \in S$ there is an edge (a_u, b_S) ,
- there are edges (u, b^*) for every $u \in \mathcal{U}$.

Thus, we have a total of $711n$ vertices and $1278n$ edges in G .

We will show the following two properties of the reduction.

- (A) A solution of 3-SET-PACKING on $(\mathcal{U}, \mathcal{S})$ with s sets correspond to a solution of LINEAR-COST-BICLIQUE-COVER on G of total cost $1278n - s$.
- (B) A solution of LINEAR-COST-BICLIQUE-COVER on G of total cost y can be transformed in polynomial time to another solution of cost $1278n - s' \leq y$ which corresponds to a solution of 3-SET-PACKING of size s' on $(\mathcal{U}, \mathcal{S})$.

Note that the above two properties prove our theorem. Indeed, if the maximum size set packing is above $(140 - \varepsilon)n$ then the solution of LINEAR-COST-BICLIQUE-COVER is of cost at most $(1138 + \varepsilon)n$. Conversely, if every set packing has size no more than $(139 + \varepsilon)n$ then all solutions of LINEAR-COST-BICLIQUE-COVER on G have a cost of at least $(1139 - \varepsilon)n$. This provides an inapproximability gap of roughly $1139/1138$.

Proving Property (A) is easy. For every set $S = \{u, v, w\}$ in the solution of 3-SET-PACKING, we take the 3×2 biclique $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ of cost 5. Note that all the selected bicliques are edge-disjoint and thus cover exactly $6s$ edges of the graph with a cost of $5s$. We cover the remaining edges individually (*i.e.*, by 1×1 biclique). Thus, the total cost is $5s + (1278n - 6s) = 1278n - s$.

Now, we turn to proving Property (B). We first observe some simple properties of G . Let a biclique $K_{p,q}$ be *trivial* if either $\min\{p, q\} = 1$ (*i.e.*, it is a star) or $p = q = 2$. Note that if a biclique is trivial, replacing it with the bicliques corresponding to its edges does not change the cost. Thus, in the sequel *we assume that all bicliques in a cover are either non-trivial or single edges*.

Lemma 8. *The only non-trivial bicliques in G are 3×2 bicliques $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ such that $S = \{u, v, w\} \in \mathcal{S}$. Moreover, any two such bicliques share at most one edge, which has b^* as one of its endpoints.*

Proof. First, note that no (trivial or non-trivial) biclique involves two vertices b_S and $b_{S'}$ for two sets $S, S' \in \mathcal{S}$ because $|S \cap S'| \leq 1$. Thus, a non-trivial biclique must involve the vertex b^* and another vertex b_S for some $S \in \mathcal{S}$. But b_S has edges only to the three vertices $a_u, a_v, a_w \in A$ such that $S = \{u, v, w\} \in \mathcal{S}$.

Next, consider two 2×3 bicliques $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ and $(\{a_{u'}, a_{v'}, a_{w'}\}, \{b_{S'}, b^*\})$ with $S = \{u, v, w\} \in \mathcal{S}$ and $S' = \{u', v', w'\} \in \mathcal{S}$. If $S \cap S' = \emptyset$ then the two bicliques are edge-disjoint. Otherwise, $|S \cap S'| = 1$ and let $v = v'$ be this common vertex; then the two bicliques share only the edge $\{a_v, b^*\}$.

Suppose that we are given a solution of LINEAR-COST-BICLIQUE-COVER of cost y on G . We normalize the solution in stages as described below. Each stage executes in polynomial time and assumes that the transformation of all previous stages have already been carried out.

- (1) If there exists a 3×2 biclique $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ which is covered individually by its edges with a cost of 6 and no edges of this biclique appear in any other bicliques, then we replace the edges by the biclique of cost 5, thereby decreasing the cost by 1. We repeat this step in arbitrary order until it does not apply anymore. Thus, after this stage, we have a maximal set of non-trivial bicliques.
- (2) Suppose now that we have two 3×2 bicliques that share an edge (a_v, b^*) . We arbitrarily replace one biclique by its individual edges excluding the edge (a_v, b^*) . We do not increase the cost and still cover all the edges. We repeat this step in any arbitrary order until it does not apply anymore.
- (3) We now observe that after (2) all the 3×2 bicliques do not share any vertex of A . Indeed, suppose that this not the case and consider two bicliques $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ and $(\{a_{u'}, a_{v'}, a_{w'}\}, \{b_{S'}, b^*\})$ with say $v = v'$. Then, they also share the edge (a_v, b^*) .
- (3) After (2), we have a solution of LINEAR-COST-BICLIQUE-COVER on G of cost $y' \leq y$ that consists only of some s' edge-disjoint 3×2 bicliques and single edges. Thus, the cost of the solution is $y' = 5s' + (1278n - 6s) = 1278n - s'$. For each 3×2 biclique $(\{a_u, a_v, a_w\}, \{b_S, b^*\})$ we select the set $\{a_u, a_v, a_w\}$ in the solution of 3-SET-PACKING on $(\mathcal{U}, \mathcal{S})$. Since no two 3×2 bicliques share a vertex of A , no two sets in the solution of 3-SET-PACKING share an element.

This concludes our proof of Property (B).

C Proof of Theorem 3

The following result is a direct consequence of results in [35, 26], providing a $2^{\log^{1-\varepsilon} n}$ -inapproximability result for MINREP under the complexity-theoretic assumption of $\text{NP} \not\subseteq \text{DTIME}(n^{\text{polylog}(n)})$.

Theorem 10. [26] *Let L be any language in NP and $0 < \varepsilon < 1$ be any constant. Then, there exists a reduction running in quasi-polynomial time, that, given an instance x of L , produces an instance M of size s of MINREP such that:*

- if $x \in L$ then M has a solution of size $\alpha + \beta$;
- if $x \notin L$ then every solution for M has size at least $(\alpha + \beta) \cdot 2^{\log^{1-\varepsilon} s}$.

C.1 Proof of Lemma 3

The variable x cannot be the head of an implicate as the truth assignment setting x to 0 and all other variables to 1 satisfies φ and falsifies such an implicate.

Clauses of the form $x \rightarrow v$, where v is a node or super-edge variable are implicates as starting from x we can mark first all node variables and then all super-edge variables. If an implicant contains another variable in its body besides x than that variable can be deleted and we still get an implicant.

C.2 Proof of Lemma 4

Consider truth assignment setting to 1 every variable in $U \cup U'$, and setting every other variable to 0. This truth assignment satisfies φ but falsifies every clause with body contained in $U \cup U'$ and head outside this set.

C.3 Proof of Lemma 5

Consider the clauses in ψ containing x , and let the set of their heads be $U_1 \cup U_2$, where $U_1 \subseteq \mathcal{V}_{\text{node}}$ and $U_2 \subseteq \mathcal{V}_{\text{super-edge}}$. For every $e \in U_2$, pick node variables u, v such that $u, v \rightarrow e$ is a witness clause, and replace $x \rightarrow e$ by the prime implicates $x \rightarrow u, x \rightarrow v$ and $u, v \rightarrow e$ in ψ . The new formula ψ' is still equivalent to φ . Every clause in ψ' containing x has a node variable head.

Assume that $U = \{u : u \in \mathcal{V}_{\text{node}}, (x \rightarrow u) \in \psi\}$ is not a solution to MINREP and consider a node variable $v \notin U$. As $x \rightarrow v$ is a prime implicate of ψ' , the forward chaining procedure over ψ' , starting from x , has to mark v . The procedure, first, can mark all node variables in U . As U is not a solution to MINREP, Lemma 4 implies that after that we can mark all super-edges variables witnessed by U and nothing else. Hence v cannot be marked. This implies that U must be a solution to MINREP, and so $|U| \geq \text{OPT}(M)$. But $|U| \leq |U_1| + 3|U_2| \leq 3(|U_1| + |U_2|)$, and so the lemma follows.

C.4 Proof of Lemma 6

To prove part (a), suppose that there is a solution U of MINREP containing α nodes of A and β nodes of B (and, thus of size $\alpha + \beta$). Then the formula ψ consisting of all witness and feedback clauses, plus all amplification clauses with heads in U , is equivalent to φ . This formula has the required size. In order to show that it is equivalent to φ , we only need to show that implies all amplification clauses $x \rightarrow u$ with $u \notin U$. Using ψ , starting from x , we can first mark U . Then, as U is a solution to MINREP, we can mark every super-edge variable using witness clauses, and, finally, we can mark u using the feedback clause with head u .

To prove part (b), suppose that $\text{OPT}(G) \geq (\alpha + \beta) \cdot 2^{\log^{1-\epsilon} s}$ and consider any Horn formula ψ equivalent to φ . Lemma 5 implies that every amplification variables must participate in at least $\text{OPT}(M)/3$ clauses and since there are t such variables ψ contains at least $t(\alpha + \beta) \cdot (2^{\log^{1-\epsilon} s}/3)$ amplification clauses. Thus, $\text{OPT}(\varphi) \geq t(\alpha + \beta) \cdot (2^{\log^{1-\epsilon} s}/3)$.

C.5 Completing the proof

We are now ready to finish the proof of Theorem 3. Set $t = s^3$ and remember that $m < s^2$. Let L be any language in NP and x be an input to L .

- If $x \in L$, then $\text{OPT}(M) = \alpha + \beta$ which implies $\text{OPT}(\varphi) \leq s^3 \cdot (\alpha + \beta) + m + s \leq s^3 \cdot (\alpha + \beta) \cdot (1 + o(1))$.
- If $x \notin L$, then $\text{OPT}(M) \geq (\alpha + \beta) \cdot 2^{\log^{1-\varepsilon} s}$ which implies $\text{OPT}(\varphi) \geq s^3(\alpha + \beta) \cdot (2^{\log^{1-\varepsilon} s} / 3)$.

This gives an inapproximability bound of $2^{\log^{1-\varepsilon} s}$ for any $0 < \varepsilon < 1$. Since the number of variables n is given by $n = s^3 + s + m < 3s^3$, this gives us an inapproximability bound of $2^{\log^{1-\varepsilon} n}$.

D Proof of Theorem 4

The problem is clearly in co-NP. For its co-NP-completeness, a reduction is given from the problem of 2-coloring 3-uniform hypergraphs, proved to be NP-complete by Lovász [31]: given $H = (S, E)$, where S is a set and E is a set of 3-element subsets of S , can the elements of S be assigned the colors *red* and *blue* such that no triple in E is monochromatic?

Let $S = \{a_1, \dots, a_n\}$, $E = \{e_1, \dots, e_m\}$ and $e_i = \{a_{i,1}, a_{i,2}, a_{i,3}\}$.

Consider the set of variables $X = \{x_1, \dots, x_n, u\}$, and let $\{y_1, \dots, y_m, z_1, \dots, z_m\}$ be additional variables. Let

$$\varphi = \left(\bigwedge_{i=1}^m \bigwedge_{j=1}^3 (x_{i,j} \rightarrow y_i) \right) \wedge (y_1, \dots, y_m \rightarrow u) \quad (3)$$

and

$$\psi = \left(\bigwedge_{i=1}^m (x_{i,1}, x_{i,2}, x_{i,3} \rightarrow z_i) \right) \wedge \left(\bigwedge_{i=1}^m \bigwedge_{j=1}^m (x_{i,j} \rightarrow y_i) \right) \wedge \left(\bigwedge_{i=1}^m (z_i, y_1, \dots, y_m \rightarrow u) \right). \quad (4)$$

We claim that H is 3-colorable iff φ and ψ are not X -equivalent.

For indices $\mathbf{j} = (j_1, \dots, j_m)$, where $1 \leq j_i \leq 3$, let $C_{\mathbf{j}}$ the clause

$$C_{\mathbf{j}} = (x_{1,j_1}, \dots, x_{m,j_m} \rightarrow u).$$

Considering all possible resolutions in φ it follows that the prime implicates of φ containing only variables from X are precisely the clauses of the form $C_{\mathbf{j}}$.

Assume that H is 2-colorable, and let a_{i,j_i} be a red element of e_i , for $i = 1, \dots, m$. Consider the truth assignment which makes variables x_{i,j_i} and y_i ($i = 1, \dots, m$) true and all other variables false. This truth assignment satisfies ψ but falsifies $C_{\mathbf{j}}$ for $\mathbf{j} = (j_1, \dots, j_m)$. Thus $C_{\mathbf{j}}$ is not an implicate of ψ .

Assume now that H is not 2-colorable, and consider a clause $C_{\mathbf{j}}$ for $\mathbf{j} = (j_1, \dots, j_m)$. We claim that $C_{\mathbf{j}}$ is an implicate of ψ . Consider the coloring of S

where elements a_{i,j_i} are red. Then, as H is not 2-colorable, it must be the case that some edge e_i contains three red elements. The clause C_j is obtained by resolving the clauses $(x_{i,1}, x_{i,2}, x_{i,3} \rightarrow z_i), (x_{i,j_i} \rightarrow y_i)$ with $(z_i, y_1, \dots, y_m \rightarrow u)$. It also has to be shown that ψ has no other prime implicants besides the clauses C_j . This follows by noting that the only way to ‘resolve away’ the y and z variables is to use some clause $(x_{i,1}, x_{i,2}, x_{i,3} \rightarrow z_i)$, some clauses $(x_{i,j_i} \rightarrow y_i)$ for every i and the clause $(z_i, y_1, \dots, y_m \rightarrow u)$. All clauses obtained this way are subsumed by some clause of the form C_j .

E Proof of Proposition 1

All prime implicants of a CNF can be generated by the variable depletion procedure or Tison’s algorithm [39]. This procedure works by processing the variables in some order. For each variable, all resolutions using that variable are performed, and clauses subsumed by other clauses are removed.

Apply the variable depletion procedure to φ starting with variables not in X , and let ψ consist of clauses over X after all variables not in X have been processed. Then ψ is X -equivalent to φ .

The definition of Steiner equivalence implies that the depletions of different variables are independent from each other. The same result is obtained if the depletions are performed in any order, or even in parallel. Also, no new bodies are formed: every body occurring in the depleted formula occurs in the Steiner extension as well.

If we deplete t variables, and the i ’th variables occurs c_i times as a head and d_i times as a body, then the depletion of that variables creates $c_i \cdot d_i$ clauses. This implies the size bound given.

F Proof of Corollary 1

The Steiner X -equivalence of φ_1 and φ_2 can be decided by using Proposition 1 to produce formulas $\psi_1(X)$ and $\psi_2(X)$ and checking their equivalence.

G Proof of Theorem 6

We are going to consider different notions of optimality as summarized in the following definition.

Definition 5. (*Notions of optimality*) For a Horn formula φ over the set of variables X , let

- $OPT(\varphi)$ denote the minimal number of clauses in any Horn formula equivalent to φ ,
- $OPT_S(\varphi)$ the minimal number of terms in any Horn formula Steiner X -equivalent to φ ,

- $OPT - BODY(\varphi)$ the minimal number of distinct bodies in any Horn formula equivalent to φ ,
- $OPT - BODY_S(\varphi)$ the minimal number of distinct bodies containing only variables from X in any Horn formula Steiner X -equivalent to φ .

There are various relationships between these notions such as the one stated below.

Proposition 2.

$$OPT - BODY_S(\varphi) = OPT - BODY(\varphi) \leq OPT_S(\varphi) \leq OPT(\varphi) \leq \frac{1}{4}OPT_S^2(\varphi)$$

The first and last relationships follow from the proof of Proposition 1 and the other follow directly from the definitions.

G.1 Proof of Lemma 7

The algorithm, referred to below as $\sqrt{\log n} - HORN - MIN$, works as follows. Given a Horn formula φ , it starts listing its prime implicates, using the algorithm of [9]. It is shown in [12] that a k -term CNF can have at most $2^k - 1$ prime implicates. If more than $2^{\sqrt{\log n}}$ prime implicates are produced, or the running time allowed for $2^{\sqrt{\log n}}$ prime implicates is exceeded, the procedure is terminated and 'none' is output. Otherwise we know that φ has at most $2^{\sqrt{\log n}}$ prime implicates. Then all sets of at most $\sqrt{\log n}$ prime implicates are checked for equivalence with φ , ordered according to increasing size, and the first one equivalent to φ is output. If neither of these is equivalent then 'none' is output. This algorithm clearly works, and as

$$\binom{2^{\sqrt{\log n}}}{\sqrt{\log n}}$$

is polynomial in n , its running time is polynomial.

G.2 Completing the proof

Given a Horn formula φ over X , let $ALG(\varphi)$ be the number of clauses in the formula produced by the algorithm.

If $\sqrt{\log n}$ -HORN-MIN(χ) is successful, then $OPT(\varphi) \leq \sqrt{\log n}$ and the formula ψ has size $OPT(\varphi) \leq (1/4)OPT_S(\varphi)^2$ by Proposition 2. Thus $\frac{ALG(\varphi)}{OPT_S(\varphi)} \leq \frac{1}{4} \cdot OPT_S(\varphi) = O(\sqrt{\log n})$.

Otherwise it holds that $OPT(\varphi) \geq \sqrt{\log n}$, and so by Proposition 2 $OPT_S(\varphi) \geq 2(\log n)^{1/4}$. Proposition 2 also implies that $OPT_S(\varphi) \geq k$. We distinguish two cases.

If $k \geq \ell$ then

$$\frac{ALG(\varphi)}{OPT_S(\varphi)} \leq c \cdot \frac{\frac{k\ell}{\ln k} + k \ln \ell}{k} = c \cdot \frac{\ell}{\ln k} + \ln \ell = O\left(\frac{\ell}{\ln \ell}\right) = O\left(\frac{n}{\ln n}\right)$$

If $k < \ell$ then

$$\begin{aligned} \frac{ALG(\varphi)}{OPT_S(\varphi)} &\leq c \cdot \left(\frac{\frac{k\ell}{\ln \ell} + \ell \ln k}{OPT_S(\varphi)} \right) \leq c \cdot \left(\frac{\ell}{\ln \ell} + \frac{\ell \ln OPT_S(\varphi)}{OPT_S(\varphi)} \right) \\ &\leq c' \cdot \left(\frac{n}{\ln n} + n \cdot \frac{\ln \ln n}{(\ln n)^{1/4}} \right) = O \left(\frac{n \ln \ln n}{(\ln n)^{1/4}} \right). \end{aligned}$$

H Proof of Theorem 7

We use a modification of the gadgets in the reduction in Theorem 3 in order to get a 3-Horn formula. In addition to the variables used there, we also use *feedback* variables $f_1, \dots, f_{p-1}$. The clauses of φ are the same as in Theorem 3, except the s feedback clauses are replaced by a set of $s + p - 1$ clauses where the $p - 1$ of these clauses form an “inverted” complete binary tree using the feedback variables. Without loss of generality assume p is a power of 2. Then, the new clauses are as follows:

Feedback clauses: Define variables $f_1, \dots, f_{2p-1}$ with

$f_p = e_1, f_{p+1} = e_2, \dots, f_{2p-1} = e_p$. (Thus $f_p, \dots, f_{2p-1}$ are only used for indexing, and are not new variables.) Feedback clauses are clauses $f_1 \rightarrow u$ for every $u \in A \cup B$ and clauses $f_{2i} \wedge f_{2i+1} \rightarrow f_i$ for $1 \leq i \leq p - 1$.

The rest of the proof uses arguments similar to Theorem 3.